GSE 518
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Suggested Solutions to Problem Set 3

1. Unlike a probability mass function, a probability density function $f_{X(x)}$ is not required to take values between 0 and 1. However, a pdf $f_{X(x)}$ must satisfy $f_{X(x)} \geq 0$ for all $x \in R$ and $\int_{-\infty}^{\infty} f_{X(x)} dx = 1$. As an example you can mention an exponential random variable X with $\lambda < 1$ (e.g., see Figure 2.3.1 from CB). For example, the pdf of an exponential random variable with $\lambda = 1/2$ is $f_{X(x)} = 2e^{-2x}$ if $x \geq 0$ and zero otherwise. Clearly, when $x = 0$ we have $f_{X(x)} = 2 > 1$. Another example is the pdf of a uniform random variable x in the interval $[0, 1/2]$, whose pdf is $f(x) = 2$ if $x \in [0, 1/2]$ and zero otherwise. Here, again, $f(x) = 2 > 1$.

2.
(a) Check class notes or Theorem 1.6.5 in CB for the properties of a pdf. Note that a function is a pdf for some random variable X if and only if it satisfies those two properties. (i) Since $f(x)$ and $F(x)$ is each a valid pfd and cdf, respectively, then we know that both must be nonnegative. Therefore $g(x) \geq 0$ for all $x \in R$. (ii) We now need to show that it integrates to one.

$$\begin{aligned}
\int_{-\infty}^{\infty} g(x)\, dx &= \int_{-\infty}^{x_0} g(x)\, dx + \int_{x_0}^{\infty} g(x)\, dx \qquad \text{(the first term is 0 by def. of } g(x)\text{)} \\
&= \int_{x_0}^{\infty} \frac{f(x)}{[1 - F(x_0)]} dx \\
&= \frac{\int_{x_0}^{\infty} f(x)\, dx}{[1 - F(x_0)]} \\
&= \frac{[1 - F(x_0)]}{[1 - F(x_0)]} \\
&= 1
\end{aligned}$$

where in the first line we used the fact that the first term is zero by definition of $g(x)$, and in the fourth equality we used the fact that $\int_{x_0}^{\infty} f(x)\, dx = 1 - F(x_0)$. Therefore, since $g(x)$ satisfies properties (i) and (ii), it is a valid pdf.
(b) This is a problem similar to the one in (a), but now with a discrete random variable. A function is a pmf for some random variable X if and only if it satisfies the two properties discussed in class. (i) $x > 0$ and $k > 0$, and therefore, $f(x) = 2x/(k(k+1)) \geq 0$ for all x. (ii) We need to show that it adds to one.

$$\begin{aligned}
\sum_{x=1}^{k} \frac{2x}{k(k+1)} &= \frac{2}{k(k+1)} \sum_{x=1}^{k} x \\
&= \frac{2}{k(k+1)} \frac{k(k+1)}{2} \\
&= 1
\end{aligned}$$

where we used the fact that $\sum_{x=1}^{k} x$ is an arithmetic series which equals $k(k+1)/2$. Therefore, $f(x)$ is a valid pmf.

(c) Check class notes or Theorem 1.5.3 in CB for the properties of a cdf. Note that a function is a cdf for some random variable X if and only if it satisfies those three properties. (i) $\lim_{x\to-\infty} F_X(x) = \lim_{x\to-\infty} e^{-e^{-x}} = e^{-\infty} = 0$ and $\lim_{x\to\infty} F_X(x) = \lim_{x\to-\infty} e^{-e^{-x}} = e^{0} = 1$. (ii) Using the chain rule, $dF_X(x)/dx = e^{-x}e^{-e^{-x}} > 0$ for all x, so $F_X(x)$ is nondecreasing (in fact, it is strictly increasing). (iii) e is a continuous function, so $F_X(x)$ is also continuous and therefore right continuous.

3.

(a) We know that in order for $f(x)$ to be a valid pdf it must integrate to one.

$$\begin{aligned}\int_{-1}^{1} f(x)\,dx &= \int_{-1}^{0} 0.2dx + \int_{0}^{1} (0.2+cx)\,dx \\ &= 0.2x|_{-1}^{0} + \left(0.2x + \frac{cx^2}{2}\right)\Big|_{0}^{1} \\ &= 0.2 + 0.2 + \frac{1}{2}c = 1\end{aligned}$$

Therefore, $c = 1.2$. Note that in the first line we brake the integral in two parts because of the definition of $f(x)$ in this problem, 0.2 for $-1 \le x \le 0$ and $0.2 + cx$ for $0 < x \le 1$. See attached graph.

(b) Remember that by definition $F(x) = \int_{-\infty}^{z} f(x)\,dx$. Here we need to be careful and consider the two pieces of the pdf (see attached graph). In particular, for values $-1 \le x \le 0$ we have

$$F(x) = \int_{-1}^{x} 0.2dx = 0.2x|_{-1}^{x} = 0.2x + 0.2 \tag{1}$$

and for values $0 < x \le 1$ we have

$$\begin{aligned}F(x) &= \int_{-1}^{0} 0.2dx + \int_{0}^{x} (0.2+1.2x)\,dx \\ &= 0.2x|_{-1}^{0} + \left(0.2x + 0.6x^2\right)|_{0}^{x} \\ &= 0.2 + 0.2x + 0.6x^2 \end{aligned} \tag{2}$$

Combining equations (1) and (2) we find the cdf of x

$$F(x) = \begin{cases} 0 & \text{if} \quad x < -1 \\ 0.2x + 0.2 & \text{if} \ -1 \le x \le 0 \\ 0.2 + 0.2x + 0.6x^2 & \text{if } 0 < x \le 1 \\ 1 & \text{if} \quad 1 < x \end{cases}$$

(Remember that $F(x)$ must be defined for *all* real numbers. Thus, it is important to add the first and last rows in the definition of $F(x)$ above).

(c) We can use the cdf to calculate this probability (see attached graph):

$$\begin{aligned}\Pr\left(0 \leq X \leq 0.5\right) &= F\left(0.5\right) - F\left(0\right) \\ &= \left[0.2 + 0.2\left(0.5\right) + 0.6\left(0.5\right)^2\right] - \left[0.2\left(0\right) + 0.2\ \right] \\ &= 0.25\end{aligned}$$

4. Here we use the results on transformation of random variables. We are given a random variable X with its pdf, and a function Y of X. We can use result 3 from class notes (Theorem 2.1.5 from CB) to obtain the distribution of $Y = aX + b$ [1]. The sample space of X is $(-\infty, \infty)$, so given the function $Y = aX + b$ the sample space of Y is also $(-\infty, \infty)$. First we check that the function is strictly monotone. Note that $dY/dX = g'\left(X\right) = a$. It is fair to assume $a \neq 0$ (otherwise Y is not a function of X), so regardless of the sign of a the function is strictly monotone, and we can use result 3. Solving for X we find $X = g^{-1}\left(y\right) = \left(y - b\right)/a$, and therefore $dg^{-1}\left(y\right)/dy = 1/a$. Applying result 3 we find

$$\begin{aligned}f_Y\left(y\right) &= f_X\left(g^{-1}\left(y\right)\right)\left|\frac{dg^{-1}\left(y\right)}{dy}\right| \\ &= \left|\frac{1}{a}\right|\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{\frac{-\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right\} \\ &= \frac{1}{\left|a\right|\sqrt{2\pi\sigma^2}}\exp\left\{\frac{-\left(\frac{y-b-a\mu}{a}\right)^2}{2\sigma^2}\right\} \\ &= \frac{1}{\sqrt{2\pi\left(\sigma a\right)^2}}\exp\left\{\frac{-\left(y - \left(a\mu + b\right)\right)^2}{2\left(\sigma^2 a^2\right)}\right\} \qquad (3)\end{aligned}$$

where in the last line we used the fact that $\left|a\right| = \sqrt{a^2}$ and collected terms. Hence the pdf of Y is given by (3) for values of $y \in (-\infty, \infty)$. Note that the pdf in (3) is also the pdf of a normal distribution with mean $a\mu + b$ and variance $\sigma^2 a^2$. Therefore, Y follows a normal distribution mean $a\mu + b$ and variance $\sigma^2 a^2$, or $N\left(a\mu + b, \sigma^2 a^2\right)$.
$Z = \left(X - \mu\right)/\sigma$ is a special case of Y with $a = 1/\sigma$ and $b = \mu - b$. Therefore, using the previous result we know Z follows a normal distribution mean 0 and variance 1, or $N\left(0, 1\right)$. This distribution is called the *standard normal distribution.*

5. We start by checking the sample spaces. The sample space of X is $\{0, 1, 2, \ldots\}$. Given that $Y = X/\left(X + 1\right)$, the sample space of Y is given by $\{0, 1/2, 2/3, 3/4, 4/5, \ldots\}$. Next, as usual when working with transformations of random variables, we check if the function is strictly monotone and consequently has an inverse. Otherwise finding the distribution of the transformation gets more difficult. In this case we have that $Y = X\left(X + 1\right)^{-1}$, so that $dY/dX = \left(X + 1\right)^{-1} - X\left(X + 1\right)^{-2} = 1/\left(X + 1\right)^2 > 0$. Therefore the function is strictly monotone and we can apply result 4 from the notes.
Solving for X we find $x = g^{-1}\left(y\right) = y/\left(1 - y\right)$. Using result 4 from notes we find

[1] Note that it is not possible to get a closed form expression for the cdf of X, so we cannot use Result 2 from class notes (Theorem 2.1.3 in CB).

$$\begin{aligned} f_Y(y) &= f_X\left(g^{-1}(y)\right) \\ &= \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^{\frac{y}{1-y}} \end{aligned}$$

Therefore, the pmf of Y is given by

$$f_Y(y) = \begin{cases} \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^{y/(1-y)} & \text{for } y = 0, 1/2, 2/3, 3/4, 4/5, \ldots, \frac{x}{1+x} \\ 0 & \text{otherwise} \end{cases}$$

6. In all the problems below we follow the same approach as in question 4: (i) get sample space, (ii) check the function is strictly monotone in the relevant space, (iii) get $g^{-1}(y)$ and $dg^{-1}(y)/dy$, (iv) apply result 3 from the class notes.
(b) The sample space of X is $(0, \infty)$, so given that $Y = 4X + 3$, the sample space of Y is $(3, \infty)$. In this case $dY/dX = 4 > 0$, so the function is strictly increasing and we can use result 3 from class notes. In this case we have $X = g^{-1}(y) = (y-3)/4$, and therefore $dg^{-1}(y)/dy = 1/4$. Using the "change of variable formula" we find

$$f_Y(y) = f_X\left(g^{-1}(y)\right)\left|\frac{dg^{-1}(y)}{dy}\right| = 7e^{-7(y-3)/4}\left(\frac{1}{4}\right) = \frac{7}{4}e^{-7(y-3)/4} \text{ for } 3 < y < \infty$$

The pdf integrates to one

$$\int_3^\infty \frac{7}{4}e^{-7(y-3)/4}dy = -e^{-7(y-3)/4}|_3^\infty = 0 - (-1) = 1$$

(c) The sample space of X is $(0, 1)$, so given $Y = X^2$, the sample space of Y is $(0, 1)$. In this case $dY/dX = 2X > 0$ for $X \in (0, 1)$, so the function is strictly increasing in the relevant space (i.e., where $f_X(x) > 0$) and we can use result 3 from class notes. Note that the function does not need to be strictly monotone everywhere to use Result 3 from class notes, it just has to be strictly increasing in the sample space of X. In this case we have $X = g^{-1}(y) = \sqrt{y}$, and therefore $dg^{-1}(y)/dy = \frac{1}{2}y^{-1/2}$. Using the "change of variable formula" we find

$$f_Y(y) = f_X\left(g^{-1}(y)\right)\left|\frac{dg^{-1}(y)}{dy}\right| = 30y\left(1 - y^{1/2}\right)^2\left(\frac{1}{2}y^{-1/2}\right) = 15y^{1/2}\left(1 - y^{1/2}\right)^2, \text{ for } 0 < y < 1$$

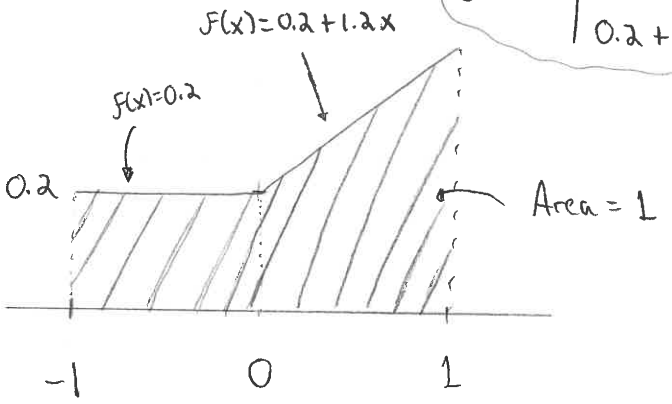
The pdf integrates to one

$$\begin{aligned} \int_0^1 15y^{1/2}\left(1 - y^{1/2}\right)^2 dy &= \int_0^1 \left(15y^{1/2} - 30y + 15y^{3/2}\right) dy \\ &= 15\left(\frac{2}{3}\right) - 30\left(\frac{1}{2}\right) + 15\left(\frac{2}{5}\right) \\ &= 1 \end{aligned}$$

Figures for Question 3:

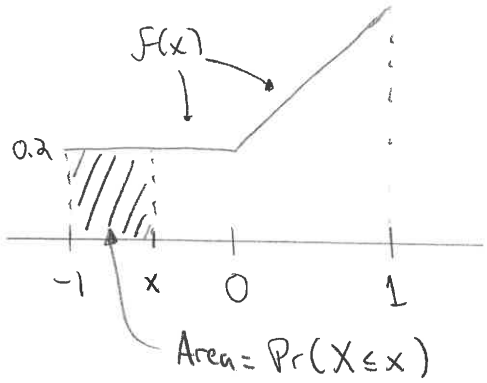
$$F(x) = \begin{cases} 0.2 & \text{if } -1 \le x \le 0 \\ 0.2 + 1.2x & \text{if } 0 < x \le 1 \end{cases}$$

3(a)

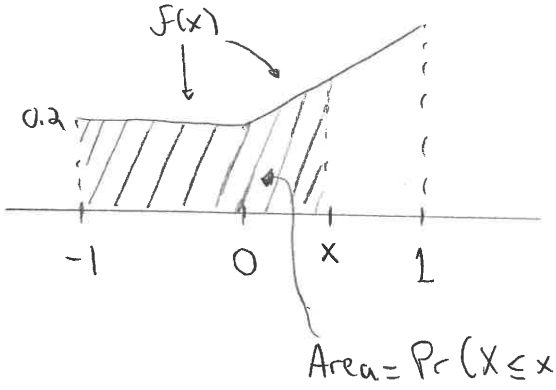


3(b)

- For given x such that $-1 \le x \le 0$:



- For given x such that $0 < x \le 1$:



3(c)

$Pr(0 \le X \le 0.5)$

